

Mekong Lens: Interactive Water-Risk Visualization for the Vietnamese Mekong Delta

Tran Anh Chuong
[24chuong.ta@vinuni.edu.vn]

Nguyen Tran Nhat Minh
[24minh.ntn@vinuni.edu.vn]

Duong Hien Chi Kien
[24kien.dhc@vinuni.edu.vn]

Phung Ba Quan
[23quan.pb@vinuni.edu.vn]

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GitHub: <https://github.com/jimmytrn154/COMP4010-Project-2>

Abstract

Mekong Lens is an interactive Python Shiny dashboard for exploring province-level water-related patterns in the Vietnamese Mekong Delta. The system combines rainfall, rainfall anomaly, dryness indicators, surface-water extent, population exposure, cropland exposure, daily rainfall-derived features, and rainfall forecast outputs into a visual analytics workflow. Rather than serving as an official flood-warning system, Mekong Lens emphasizes transparent exploratory indicators such as rainfall anomaly, rainfall z-score, dry-day ratio, maximum consecutive dry days, water area, and water-area percentage. The dashboard uses linked maps, KPI cards, rankings, trend charts, heatmaps, province comparisons, and a machine learning forecast preview to help users understand how wet, dry, and surface-water signals vary across provinces and seasons.

1 Introduction

1.1 Background and Motivation

The Vietnamese Mekong Delta is one of Vietnam's most important agricultural and livelihood regions. It supports rice production, aquaculture, transportation, and millions of residents. At the same time, the region is highly exposed to changing water conditions, including abnormal rainfall, seasonal flooding, prolonged dryness, and surface-water variation. These conditions are difficult to interpret from raw satellite or geospatial datasets because they vary across provinces, months, and years.

Mekong Lens addresses this challenge through an interactive visual analytics dashboard. Instead of presenting a static chart or a single summary value, the dashboard allows users to explore multiple water-related indicators across the 13 Mekong Delta provinces. The project focuses on province-level patterns because this scale is detailed enough for regional comparison while still readable in an interactive dashboard.

1.2 Central Question

The central question of the project is:

How do rainfall, dryness, surface-water extent, and exposure-related indicators vary across Mekong Delta provinces over time, and which provinces show stronger wet, dry, or surface-water signals under selected conditions?

This question reflects the current direction of the project. Mekong Lens does not try to produce an official disaster warning. Instead, it supports exploratory analysis by showing interpretable indicators that users can inspect directly.

1.3 Project Goal

The goal of Mekong Lens is to transform multi-source environmental and geospatial data into a coherent interactive data story. Users can begin with a regional overview, then move into province-level maps, rainfall and dryness trends, surface-water patterns, province comparisons, and rainfall forecast previews. This structure helps users understand not only where strong signals appear, but also whether those signals are seasonal, unusual, persistent, or exposure-relevant.

The project is designed with responsible interpretation in mind. Surface-water extent is treated as a flood or wetness proxy, not a direct measurement of flood damage. The machine learning tab is framed as a rainfall forecast preview, not an official advisory. This avoids overclaiming while still allowing meaningful exploration.

2 Data Sources and Preprocessing

2.1 Data Sources

Mekong Lens uses public satellite and geospatial datasets collected mainly through Google Earth Engine and then processed with Python. The main analytical structure is a province-month panel, where each row represents one province in one month. This structure allows the dashboard to synchronize maps, rankings, trends, heatmaps, and comparisons using the same year, month, province, and metric filters.

2.2 Collection and Panel Construction

The data pipeline filters satellite and geospatial datasets to the Vietnamese Mekong Delta, clips rasters to province boundaries, aggregates values by province, and exports CSV or GeoJSON outputs. These files are cleaned in Python and merged into dashboard-ready tables. Data crawling is supported by Google Earth Engine.

The final dashboard panel, `province_month_panel.csv`, combines rainfall, surface water, exposure context, and daily-derived rainfall indicators. Population is merged

Table 1: Core datasets used in Mekong Lens.

Dataset	Size / Shape	Coverage	Resolution	Role in Dashboard
CHIRPS monthly rainfall	0.34 MB; 7,020 × 5	1981–2025	Province-month	Main rainfall totals, long-run monthly mean, rainfall anomaly, and rainfall z-score.
CHIRPS daily rainfall	9.22 MB; 213,668 × 6	1981–2025	Province-day	Daily rainfall structure, extreme rainfall features, dry/wet persistence indicators, and ML forecast input.
JRC Global Surface Water	0.42 MB; 5,902 × 7	1984–2021	Province-month	Surface-water area and percentage as a flood/wetness proxy.
FAO GAUL boundary	0.89 MB; 13 features	Static	Province geometry	Administrative boundaries for choropleth maps and spatial overlays.
Processed monthly panel	1.78 MB; 7,020 × 27	1981–2025	Province-month	Integrated dashboard dataset combining rainfall, water, exposure, and daily-derived rainfall features.

by province and year, while cropland is merged as a static province-level exposure layer. This merged panel supports the main dashboard views and keeps the dashboard logic consistent across tabs. The project uses four datasets: monthly rainfall for 1981–2025, daily rainfall for 1981–2025, GeoJSON boundaries for 13 Mekong Delta provinces, and the processed monthly panel. The monthly rainfall and processed panel datasets each contain 7,020 balanced province-month observations. The daily rainfall dataset contains 213,668 province-day records, or 16,436 daily timestamps for each province. Rainfall is complete for 1981–2025, while surface water covers 1984–2021, population covers 2000–2020, and cropland is available for 2021 only.

2.3 Rainfall Processing

The rainfall pipeline uses two raw rainfall files for different purposes. The monthly rainfall file is used to build the baseline rainfall indicators. It is cleaned and standardized, then used to compute each province’s long-run calendar-month climatology. From this, the pipeline derives:

- **rainfall_mm**: monthly rainfall total,
- **monthly_mean**: long-run average for the same province and month,
- **rainfall_anomaly**: deviation from the province-month normal,
- **rainfall_zscore**: standardized anomaly.

These indicators allow the dashboard to distinguish seasonal rainfall from unusually wet or dry months.

2.4 Daily-Derived Features

Monthly rainfall totals alone can hide important short-term behavior. For example, two provinces may receive similar monthly rainfall, but one may experience rainfall concentrated in a few heavy events while another may have more evenly distributed rainfall. To capture this, the daily rainfall file is aggregated into province-month

features such as:

- **max_1day_rainfall**,
- **rain_days_count** and **dry_days_count**,
- **heavy_rain_days_20mm** and **heavy_rain_days_50mm**,
- **max_consecutive_dry_days** and **max_consecutive_wet_days**,
- **rain_day_ratio** and **dry_day_ratio**.

These daily-derived features support the dashboard’s dryness interpretation and add analytical depth beyond simple monthly totals.

2.5 Surface Water and Exposure Processing

Surface-water data is processed into **water_area_km2** and **water_area_pct**. These indicators show how much of each province is mapped as surface water in a selected month. They are interpreted as a wetness or flood-related proxy rather than direct flood damage.

Population and cropland data provide exposure context. Population helps identify where water-related signals may affect more people, while cropland helps identify where agricultural exposure may be higher. Because population is annual and cropland is static, the dashboard treats them as contextual indicators rather than monthly dynamic measurements.

2.6 Data Limitations

The datasets have different strengths and limitations. CHIRPS is satellite-derived and should not be treated as exact station measurement for every local area. JRC surface water detects mapped water extent but does not measure flood depth, damage, or official flood impact. Population is annual rather than monthly, and cropland is based on a static 2021 land-cover snapshot. These differences motivate the dashboard’s coverage-aware design: users should see unavailable or limited variables honestly rather than being given misleading values.

Table 2: Main dashboard components.

Component	Purpose
Sidebar filters	Select year, month, province, Top-N provinces, and metric.
KPI cards	Summarize selected-period conditions quickly.
Choropleth map	Locate province-level rainfall, dryness, or surface-water patterns.
Ranking chart	Compare provinces under the selected indicator.
Line chart	Show seasonal and historical rainfall behavior.
Anomaly heatmap	Detect unusual province-month patterns.
Comparison panel	Compare selected province with the wider delta.
ML forecast view	Separate forecast preview from historical observation.

3 Dashboard Design and Interaction

3.1 Visual Story Flow

The dashboard is designed as a guided but flexible analytical story. The user starts with the *Mekong Summary* tab, which provides a high-level snapshot of the selected month. KPI cards summarize the main regional conditions, while the map shows where the strongest spatial signals appear. From there, users can move into more specific tabs for rainfall and dryness, surface water, province comparison, machine learning prediction, and methodology.

The dashboard uses tabs to reduce cognitive load. Each tab has a clear analytical role: summary, rainfall/dryness, surface water, province comparison, ML prediction, and methodology/caveats. This keeps the interface readable while still allowing deeper exploration.

3.2 Main Visual Components

Mekong Lens uses multiple visualization types to support spatial, temporal, and comparative analysis. The map is the main spatial entry point. Ranking charts show which provinces stand out under the selected metric. Trend charts show seasonal or historical changes. Heatmaps reveal province-month anomaly patterns. Comparison views help users inspect one province against the regional context.

This set of components satisfies the requirement for at least five charts and at least three chart types. More importantly, each chart supports a specific analytical question rather than simply adding visual variety.

3.3 Interaction Design

The main interaction mechanism is a persistent sidebar containing global filters. Users can select a year, month, province, Top-N ranking size, and focus metric. These filters update the KPI cards, map, rankings, trend charts, heatmaps, and comparison views together. This linked-view design helps users avoid manually matching

separate charts.

Tooltips provide additional detail without overcrowding the page. On maps and charts, users can inspect province name, rainfall, rainfall anomaly, rainfall z-score, dry-day ratio, maximum dry spell, surface-water area, population, and cropland context. This keeps the layout clean while still allowing detailed exploration.

3.4 Design Justification

The dashboard avoids relying on one single score because water-related risk is multidimensional. Rainfall alone does not fully explain flood or drought conditions. Surface-water extent, dryness persistence, anomaly behavior, population, and cropland exposure all provide additional context. Therefore, the dashboard uses a linked multi-view design: users can see the regional summary, locate spatial hotspots, compare provinces, inspect trends, and read caveats in one coherent workflow. The dark card-based layout also supports readability. KPI cards highlight the most important values first, while maps and charts are separated into visual sections. This makes the dashboard feel like an analytical control panel rather than a static report.

3.5 Coverage-Aware Interaction

Coverage-aware interaction is important because the datasets do not share the same time period. Rainfall covers the longest period, surface-water data ends earlier, population is annual, and cropland is static. The dashboard should therefore communicate missing or unavailable values through caveat notes, empty states, or careful labels. This prevents users from assuming that every variable is available for every selected year.

4 Analytical and ML Methods

4.1 Spatial and Spatio-Temporal Analysis

The analytical backbone of Mekong Lens is province-level spatial and temporal comparison. Spatially, choropleth maps show how rainfall, dryness, and surface-water indicators vary across the 13 Mekong Delta provinces. Temporally, trend charts and heatmaps show how these indicators change across months and years.

Rainfall anomaly and z-score are especially useful because they compare each province to its own normal seasonal pattern. This helps avoid misleading comparisons between naturally wetter and naturally drier provinces. Anomaly heatmaps then reveal province-month combinations that are unusually wet or dry relative to historical conditions.

4.2 Daily Extreme and Persistence Indicators

Daily-derived features add an important analytical layer. Heavy-rain day counts and maximum one-day rainfall help identify months shaped by intense rainfall events. Dry-day ratio and maximum consecutive dry days help identify persistence of dry conditions. These features make the dashboard more informative than a simple

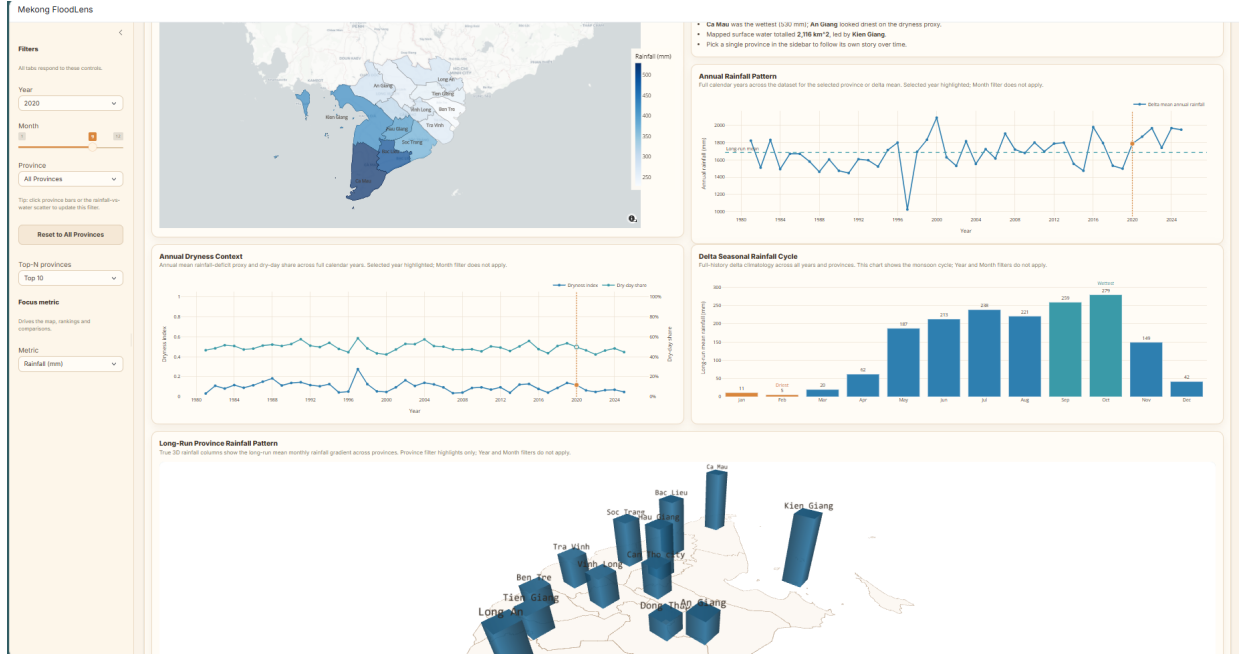


Figure 1: Main dashboard overview. This figure should show how Mekong Lens moves users from filters and summary indicators to spatial interpretation.

monthly rainfall viewer.

4.3 Machine Learning Forecasting

The ML component is a province-level daily rainfall forecasting workflow. The daily rainfall file is split into one file per province, and each province is modeled separately. The forecasting task is divided into two stages:

- an XGBoost classifier predicts whether rainfall occurs,
- an XGBoost regressor predicts rainfall amount.

The models use lag features, rolling statistics, rain-day counts, wet/dry streak features, and seasonal encodings. Forecasts are generated recursively, meaning each predicted day becomes part of the history used to build features for the next day. This approach allows the model to carry forward recent wet or dry conditions during the forecast period.

4.4 Role of ML in the Dashboard

The ML output is precomputed and loaded into the dashboard rather than trained at runtime. This keeps the Shiny application responsive and reproducible. The forecast tab is separated from the historical observation tabs so that users can clearly distinguish measured past conditions from model-generated future estimates.

The forecast is best interpreted as a rainfall regime preview. It helps users compare expected wet/dry continuation across provinces, but it should not be used to claim precise future flood events. This framing keeps the predictive component useful while avoiding overstatement.

Table 3: Technology stack.

Technology	Role
Python Shiny	Reactive dashboard framework.
Pandas	Cleaning, aggregation, feature engineering.
GeoPandas	Province boundary and geospatial data handling.
Plotly	Interactive maps, charts, heatmaps, and tooltips.
XGBoost / scikit-learn	Rainfall occurrence and amount forecasting.
Google Earth Engine	Source extraction and province-level aggregation.

5 Implementation and Reproducibility

5.1 Technology Stack

Mekong Lens is implemented primarily in Python. The dashboard uses Python Shiny for reactive application structure, Plotly for interactive visualizations, Pandas for data cleaning and feature construction, GeoPandas for boundary and geospatial handling, and XGBoost/scikit-learn for forecasting.

5.2 System Architecture

The system has three main layers. The data layer stores raw files, cleaned files, and processed dashboard tables. The dashboard layer loads the processed panel and boundary GeoJSON, then uses reactive filters to update all visual outputs. The ML layer trains outside the dashboard and writes forecast results to separate files that the app can load.

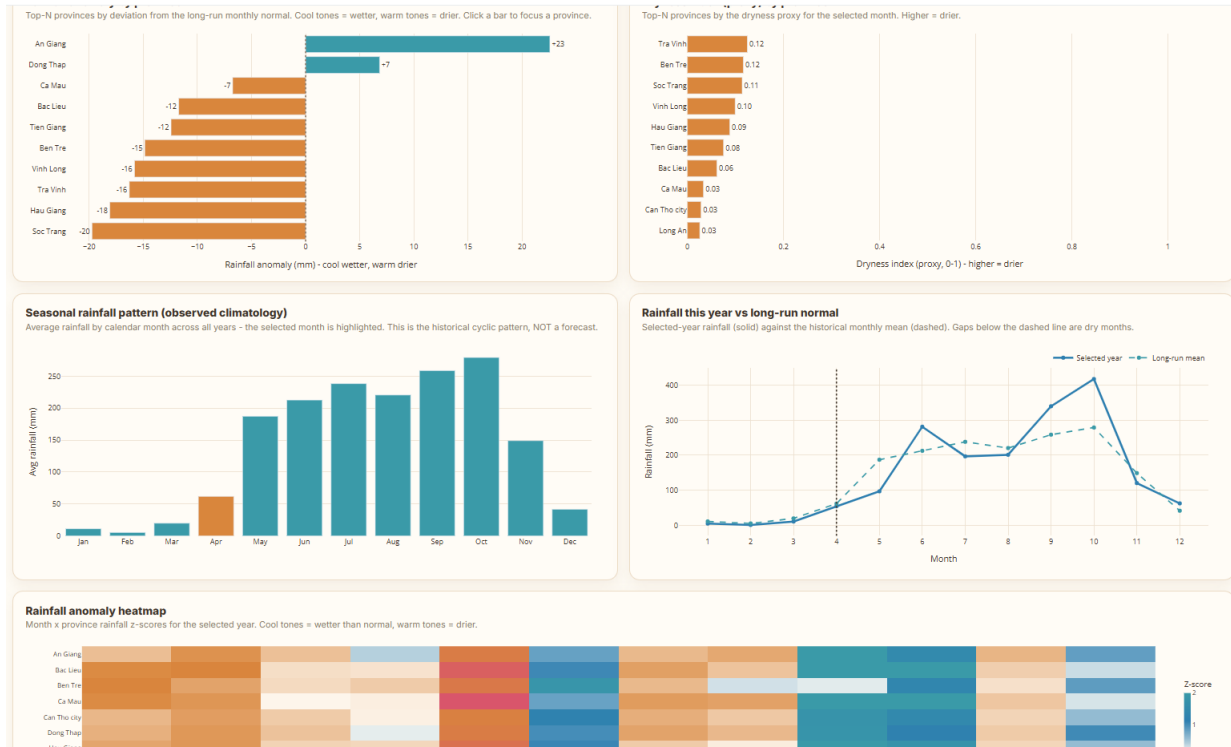


Figure 2: Rainfall and dryness view. This figure highlights how monthly and daily-derived features are combined.

6 Key Findings and Insights

6.1 Seasonal Rainfall Pattern

The processed panel shows a strong seasonal rainfall cycle across the Mekong Delta. The wettest months are October, with about 279 mm on average, and September, with about 259 mm. The driest months are February, with about 5 mm, and January, with about 11 mm. This confirms a clear monsoon-driven pattern and justifies the dashboard's focus on monthly and seasonal filtering.

6.2 Spatial Rainfall Differences

Rainfall is not evenly distributed across provinces. Coastal and western provinces are generally wetter, with the highest long-run average monthly rainfall in Ca Mau, Kien Giang, and Bac Lieu. Lower long-run averages appear in Dong Thap, Ben Tre, and An Giang. This supports the use of choropleth maps because geography is central to the water-risk story.

6.3 Dryness and Daily Persistence

Dryness indicators reveal patterns that monthly rainfall totals alone can miss. Tra Vinh, Ben Tre, and Vinh Long show higher dry-day shares and longer dry spells, while Kien Giang and Ca Mau are relatively less dry on average. This shows the value of daily-derived features: a month can have moderate total rainfall but still contain long dry spells.

6.4 Extreme Rainfall Signals

Extreme rainfall signals concentrate in the wet season, especially from May to October. Heavy-rain days above 20 mm peak around October, while maximum one-day

rainfall is strongest around September and October. Ca Mau, Bac Lieu, Soc Trang, and Kien Giang show stronger wet-season intensity signals.

6.5 Surface Water Context

Surface-water extent does not move in a simple same-month linear relationship with rainfall anomaly. This suggests that mapped surface water is influenced by more than current-month rainfall, including storage, hydrology, floodplain behavior, and timing effects. Therefore, surface water is best interpreted as a wetness or flood-related proxy rather than a direct flood impact measure.

7 Challenges and Limitations

7.1 Data Coverage Mismatch

A major limitation is that the datasets do not cover the same time range. Rainfall and daily rainfall features cover the longest period, JRC surface-water data ends in 2021, population is annual, and cropland is a static 2021 snapshot. This means not every indicator is available for every year or month. Mekong Lens handles this by presenting the dashboard as exploratory and by making coverage limitations visible in the methodology and caveat sections.

7.2 Proxy-Based Indicators

Several variables are proxies rather than direct impact measurements. CHIRPS rainfall is satellite-derived and gauge-adjusted, but it is not the same as a local station reading for every location. JRC surface-water extent shows detected water area, not flood depth, damage, or economic loss. Dryness indicators are relative analytical metrics, not official drought classifications. These lim-

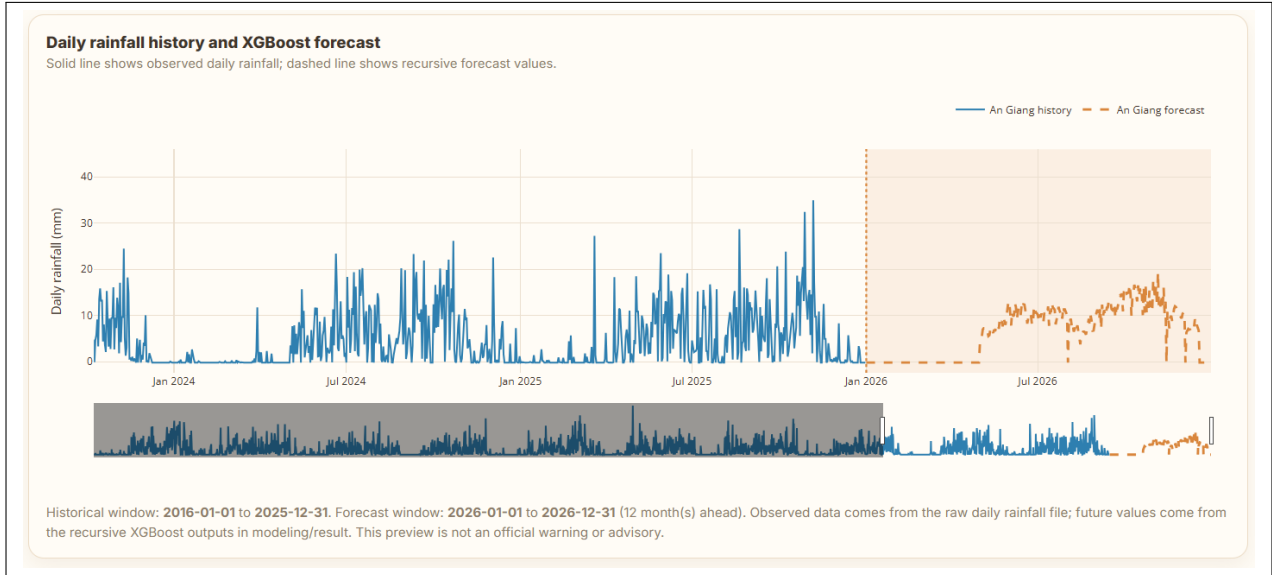


Figure 3: ML prediction tab. Forecasts are shown as rainfall regime previews rather than official flood predictions.

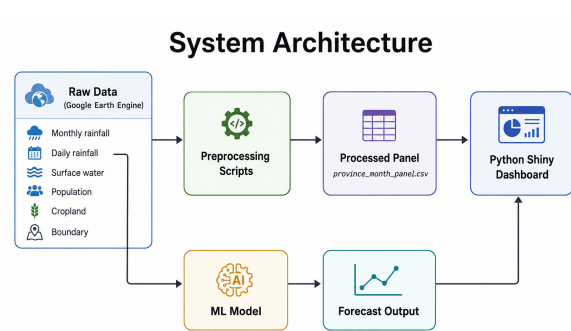


Figure 4: System architecture and data flow.

itations are why the dashboard avoids official warning language.

7.3 Spatial Aggregation

Province-level aggregation makes the dashboard readable and suitable for regional comparison, but it hides smaller-scale variation. Flooding, dryness, and surface-water extent can vary within a province, especially between districts, communes, coastal zones, riverine zones, and inland agricultural areas. Therefore, Mekong Lens is useful for broad spatial analysis but should not replace district-level or field-based assessments.

7.4 Dashboard Complexity

The dashboard combines multiple datasets, maps, rankings, heatmaps, trends, comparison views, and forecast outputs. This improves analytical depth but also increases implementation complexity. Large time-series files and geospatial rendering can affect loading time or responsiveness, so efficient preprocessing and precomputed forecast outputs are important design decisions.

8 Future Work and Conclusion

8.1 Future Work

Future work can improve Mekong Lens in four main directions. First, the ML pipeline could include more climate predictors, such as temperature, humidity, wind, or large-scale ocean-atmosphere indicators. The current model relies mainly on historical rainfall patterns, so additional climate drivers may improve inter-annual forecast realism.

Second, uncertainty visualization should be added. Forecast confidence intervals or prediction bands would help users understand that forecasts are estimates rather than deterministic outcomes. Third, district-level or finer spatial analysis would reveal local hotspots hidden by province-level aggregation. Fourth, interpretability tools such as feature-importance plots could explain which lag, rolling, or seasonal features drive the forecast for each province.

8.2 Conclusion

Mekong Lens combines spatial, temporal, exposure, and predictive components into an interactive visual analytics dashboard for the Vietnamese Mekong Delta. The project goes beyond static visualization by linking maps, KPI cards, rankings, trends, heatmaps, province comparisons, and rainfall forecast previews. Its main contribution is not official disaster prediction, but transparent exploration of how rainfall, dryness, surface water, and exposure vary across provinces and seasons.

The final dashboard supports the project goals by providing meaningful interactivity, multiple chart types, spatial and temporal analytics, a reproducible data pipeline, and a machine learning component embedded into the visual workflow. Most importantly, it presents complex environmental data in a way that remains interpretable and responsible.